



Unveiling the spectrum of vancomycin resistance in *Staphylococcus aureus* from Hospital-acquired urinary tract infections (HA-UTI) in cardiac patients

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Abstract

Background Patients with cardiac disease are highly susceptible to hospital-acquired urinary tract infections (HA-UTI) due to immunosuppression, invasive procedures, and prolonged hospitalization. Although *Enterobacteriaceae* are frequent pathogens, the contribution of vancomycin-resistant *Staphylococcus aureus* (VRSA) remains unexplored. Among 11 known glycopeptide resistance genes (*vanA*, *vanB*, *vanD*, *vanF*, *vanI*, *vanM*, *vanC*, *vanE*, *vanG*, *vanL*, and *vanN*), *vanA* and *vanB* are most commonly reported. However, the diversity of other *van* genes and their interplay with antimicrobial resistance (AMR) and virulence determinants are poorly characterized.

Methods and results Between March 2020 and April 2022, 189 cardiac ward patients were screened for VRSA-associated HA-UTI. The presence of *van* genes, efflux pump genes (*norA*, *norB*, *sepA*, *mepA*, *mdeA*), and virulence genes (*hla*, *sea*, *icaA*, *fnbA*) was determined using singleplex PCR, correlation analysis, and qRT-PCR for expression profiling. Antimicrobial susceptibility was assessed via disk diffusion, minimum inhibitory concentration (MIC), hierarchical clustering, and multiple antibiotic resistance indexing. 55% of VRSA isolates harboured limited *van* gene diversity, with *vanA* predominant. These isolates exhibited extensive AMR, high vancomycin MICs (> 256 µg/mL), and multidrug resistance. Co-occurrence of efflux pump genes (*norA* + *norB*, *norB* + *norC*, *sepA* + *norB*) and virulence genes (*hla* + *sea*, *icaA* + *fnbA*) showed significant positive correlations. All isolates were susceptible to linezolid and chloramphenicol.

Conclusion *vanA*-mediated resistance in VRSA from HA-UTI is strongly associated with efflux pump activity and virulence gene expression. Integrative genomic and transcriptomic approaches are essential for deciphering these networks, guiding precision therapeutics, and strengthening infection control measures.

Keywords *Staphylococcus aureus* · Urinary tract infections (UTIs) · Vancomycin resistance · Cardiac care unit · Hospital acquired infections (HAIs) · Antimicrobial resistance (AMR)

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Introduction

Urinary tract infections (UTIs) are common microbial illnesses that affect the kidneys, bladder, urethra, and prostate, imposing significant morbidity and healthcare costs globally [1]. Classified as community-acquired (CA-UTIs) or hospital-acquired (HA-UTI), the prevalence of these infections is influenced by factors such as age, sex, background, onset of infection, and antimicrobial use. However, HA-UTI, particularly in vulnerable populations such as cardiac patients, remains underexplored [2].

Patients with cardiac conditions are particularly susceptible to hospital-acquired infections (HAIs) [3], including HA-UTI, owing to prolonged hospital stays, frequent exposure to invasive procedures, and immunocompromised status [4]. These factors not only increase the likelihood of infection but also exacerbate the severity of outcomes when pathogens such as vancomycin-resistant *Staphylococcus aureus* (VRSA) are involved [5]. In cardiac care units, the high prevalence of multidrug-resistant (MDR) organisms further increases this risk [6, 7]. Studies have shown that cardiac patients with HAIs often experience longer hospitalizations, increased morbidity, and higher mortality rates than the general population [8].

Most UTIs are caused by gram-negative bacteria, particularly *Escherichia coli*, which accounts for 65–90% of all UTIs [8]. Other uropathogens include *Enterococcus* spp., *Klebsiella pneumoniae*, *Citrobacter* spp., *Pseudomonas aeruginosa*, and coagulase-negative staphylococci (CoNS), with *Staphylococcus aureus* being a rare cause (0.5–6% of cases), and VRSA is even less prevalent [9].

S. aureus is a ubiquitous pathogen capable of causing diseases ranging from minor skin infections to severe systemic conditions such as pneumonia, endocarditis, and bacteremia [10]. These infections are particularly severe in patients with cardiac disease, as compromised cardiovascular function can exacerbate infection-related complications, leading to worse prognoses [11]. *S. aureus* employs a variety of virulence factors to evade the host immune response, establish colonization, and cause tissue damage [12]. The genetic flexibility of pathogens has enabled the emergence of virulent and drug-resistant strains, including methicillin-resistant *S. aureus* (MRSA), which pose significant treatment challenges [13].

Vancomycin, historically the drug of choice for MRSA treatment, has shown reduced efficacy owing to the emergence of VRSA. The first case of VRSA among patients with diabetes in the United States was reported in 2002 [14]. Since then, the World Health Organization (WHO) has recognized VRSA as a high-priority pathogen, given its potential to significantly increase global fatality rates [15, 16]. VRSA is primarily attributed to the acquisition of *van*

genes through horizontal gene transfer (HGT) and mutations. Although *vanA* and *vanB* are the most commonly studied resistance genes, the existence of 11 distinct *van* genes underscores the genetic diversity and adaptability of VRSA [17].

Despite the dominance of *vanA* and *vanB* in molecular detection protocols, studies have highlighted the potential contribution of other *van* genes to resistance. HGT within hospital environments, particularly in high-risk settings such as cardiac units, facilitates the dissemination of these genes among pathogens [18]. This genetic exchange may lead to the emergence of VRSA strains with novel resistance mechanisms, complicating infection control and treatment strategies [19, 20]. Beyond *van*-mediated resistance, other genetic and phenotypic traits further strengthen VRSA's ability to survive and cause disease in patients.

Efflux pumps and virulence determinants are increasingly recognized as co-contributors to the persistence and adaptability of VRSA in clinical settings. Efflux pumps not only mediate MDR but also enhance bacterial fitness, biofilm formation, and survival under host-imposed stress, thereby complementing *van* gene-mediated glycopeptide resistance [21]. Likewise, virulence factors such as adhesins, toxins, and immune evasion proteins facilitate colonization, immune evasion, and tissue damage in infected patients [22]. The co-occurrence of these traits with diverse *van* genes may create a synergistic advantage, enabling VRSA to cause severe, treatment-refractory infections in high-risk groups, such as cardiac patients with HA-UTI.

However, little is known about how these resistance and virulence traits intersect across the full spectrum of *van* genes, particularly beyond *vanA* and *vanB*, in VRSA isolates from cardiac patients with HA-UTI. This gap is critical, as cardiac patients represent a uniquely vulnerable population with a heightened susceptibility to HAIs and poor outcomes associated with MDR infections.

Therefore, we hypothesized that VRSA isolates from cardiac patients with HA-UTI might harbour *van* genes other than *vanA* and *vanB*. To address this, we investigated the diversity of *van* genes in VRSA isolates, evaluated the pathogenicity and virulence potential of isolates harbouring the most prevalent *van* genes, examined the co-occurrence of virulence and efflux pump resistance genes, and performed a comprehensive analysis of antibiotic susceptibility. These findings provide valuable insights into effective treatment strategies, enhanced infection control, and advanced therapeutic approaches in high-risk healthcare settings.

Materials and methods

Study design

This cross-sectional descriptive study was conducted in the cardiac ward (CW) of a tertiary-level autonomous public sector hospital specializing in cardiac diseases over a two-year period from March 2020 to April 2022. The hospital is located in Faisalabad, Pakistan, at an elevation of 184 m above sea level and is situated between 31°20' and 31°33' N and 73°13' and 72°55' W. Faisalabad is the third most populous city in Pakistan. The hospital has a capacity of 202 beds.

A total of 189 cardiac patients were initially screened for HAIs, focusing on UTIs caused by VRSA. The inclusion criteria included patients showing signs of infection either upon hospital admission or within 48–72 h post-admission, a typical timeframe for HA-UTI [23]. Only those with a urine specimen showing ≥ 10 white blood cells (WBCs) per high-power field under microscopy were included in further analysis [24, 25]. These patients were considered to have HA-UTI and were subsequently screened for VRSA.

The exclusion criteria included patients who had recently received antibiotics (within the last two weeks), those with

recurrent UTIs, urological abnormalities, ongoing antibiotic treatment, or pregnancy, as these factors could influence infection dynamics and resistance patterns. Additionally, isolates from non-urine specimens were excluded to ensure a focus on HA-UTI (Fig. 1).

Demographic data (age, sex, and residence background) and clinical data on HA-UTI (infection onset) were retrieved from patient records.

Bacterial isolates

To determine the prevalence of hospital-acquired *S. aureus* among HA-UTI cardiac patients, clean-catch midstream urine (MSU) specimens were aseptically cultured on Blood Agar (Oxoid, Basingstoke Hampshire, UK) and MacConkey Agar (Oxoid, Basingstoke Hampshire, UK) for 24 h at 37 °C under aerobic conditions. Conventional morphological methods, such as Gram staining (Oxoid, Basingstoke Hampshire, UK), were used to identify staphylococcal isolates. Gram-positive cocci in clusters found under a microscope (Olympus CX23, Tokyo, Japan) were sub-cultured on selective mannitol salt agar (Oxoid, Basingstoke, Hampshire, UK).

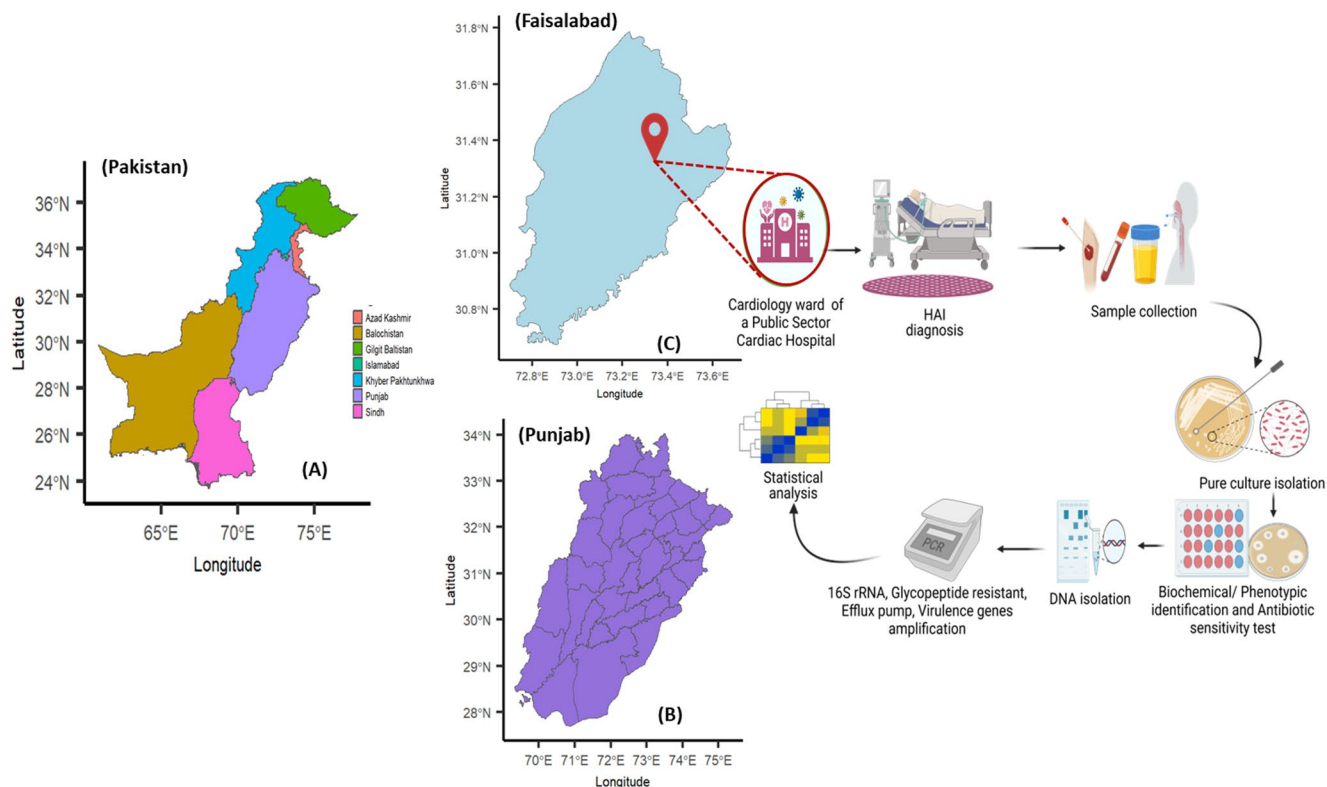


Fig. 1 Schematic Diagram of Study Design. Spatial data for the study area and comprehensive methodological details were created using R Studio and BioRender (www.biorender.com). Key: (A) Map of the Administrative Area of Pakistan (country level), (B) Map of

the Administrative Area of Punjab (province level), (C) Map of the Administrative Area of Faisalabad (district level). Abbreviations: HAI: Hospital-Acquired Infections

Moreover, several biochemical assays, including catalase (bioMérieux, Marcy L'Etoile, France), coagulase (Remel™ Coagulase Plasma, ThermoFisher Scientific, Heysham, UK), and latex agglutination tests (Slidex Staph Plus, bioMérieux, Marcy L'Etoile, France), were performed to confirm the presence of *S. aureus* [26].

Detection of VRSA isolates

To screen for vancomycin resistance among *S. aureus* isolates, each isolate was inoculated onto brain heart infusion (BHI) agar (Oxoid, Basingstoke, Hampshire, UK) that was prepared in-house. The agar contained vancomycin (Oxoid, Basingstoke, Hampshire, UK) at a concentration of 6 mg/L (BHI-V6). A 10 µL spot of each isolate with a 0.5 McFarland standard in saline was applied to BHI-V6, followed by incubation at 35 °C for 48 h under aerobic conditions. The presence of growth indicated a positive result for vancomycin resistance [27]. Isolates grown on BHI-V6 agar were further investigated to confirm their phenotypic and molecular characteristics.

Genomic DNA extraction

For molecular identification, glycopeptides encoding *van* gene diversity, resistome (efflux pump genes), and virulence analysis, genomic DNA was extracted from the VRSA strains using a commercial genomic DNA extraction kit (PureLink™ Pro 96), according to the manufacturer's instructions (ThermoFisher Scientific, Heysham, UK). The purity of the extracted DNA was evaluated using a NanoDrop 2000 spectrophotometer (ThermoFisher Scientific, USA).

16 S rRNA gene sequencing

All confirmed VRSA isolates were subjected to amplification of a partial sequence (1.5 kb fragment) of the 16 S rRNA gene using specific primers and reaction conditions (Supplementary Table 1 A). The resulting amplified products were sequenced by the 1st Base www.base.asia.com using Sanger sequencing. The resulting sequences were compared with the National Center for Biotechnology Information (NCBI) database to verify their identity.

Molecular identification and diversity of vancomycin-resistant glycopeptide-encoding Van genes

To validate the presence of phenotypically detected VRSA and assess the diversity of *van* genes, we conducted singleplex polymerase chain reaction (PCR) using

specific oligonucleotide primer sequences (Supplementary Table 1B). Each reaction consisted of 12.5 µL of AccuPrime master mix (Invitrogen, Waltham, Massachusetts, USA), 1 µL (10 pmol) of the forward primer, 1 µL (10 pmol) of the reverse primer, and 2 µL of DNA extract, and the final volume was adjusted to 25 µL with sterile H₂O, followed by amplification using an Applied Biosystems thermocycler (ThermoFisher Scientific, Heysham, UK) with the following program: initial denaturation at 94 °C for 10 min, followed by 30 cycles. Each cycle included a denaturation step at 94 °C for 30 s, an annealing step based on the *van* gene primer sets (Supplementary Table 1B) for 45 s, and an extension step at 72 °C for 30 s. After these cycles, a final extension step was performed at 72 °C for 10 min, followed by a holding step at 4 °C.

For analysis, the PCR products were electrophoresed, stained (10 µM ethidium bromide), and visualized using UV transillumination (Gel Doc EZ imager; Bio-Rad Laboratories, Hercules, California, USA).

For quality control in each test run, *Enterococcus faecium* ATCC 51,559 and *Enterococcus faecalis* ATCC 51,299 were used as positive controls for *vanA* and *vanB*, respectively. *Enterococcus faecalis* ATCC 29212 was used as a negative control for *vanA/B*.

Notably, the presence of any *van* gene among the phenotypically confirmed VRSA isolates was considered a molecular-based confirmation of VRSA isolates.

Detection of efflux pump genes

To determine the prevalence of efflux pump genes among VRSA isolates, specific oligonucleotide primer sequences for efflux pump genes (*norA*, *norB*, *norC*, *sepA*, *mepA*, and *mdeA*) were used for singleplex PCR (Supplementary Table 1 C). Each reaction consisted of 12.5 µL of AccuPrime master mix (Invitrogen, Waltham, Massachusetts, USA), 1 µL (10 pmol) of the forward primer, 1 µL (10 pmol) of the reverse primer, 2 µL of DNA extract, and the final volume was adjusted to 25 µL with sterile H₂O, followed by amplification using an Applied Biosystems thermocycler (ThermoFisher Scientific, Heysham, UK) with the following program: initial denaturation at 94 °C for 4 min, followed by 35 cycles of denaturation at 94 °C for 30 s, annealing at 60 °C (*norA*), 62 °C (*norB* and *norC*), and 61 °C (*sepA*, *mepA*, and *mdeA*) for 45–55 s, extension at 72 °C for 55 s to 1 min, and a final extension step at 72 °C for 5 min [28].

Detection of virulome genes

To assess the virulence or pathogenicity characteristics of the VRSA isolates, specific oligonucleotide primer sequences were used to detect the virulence factor genes

using singleplex PCR (Supplementary Table 1D). This included the detection of genes such as *hla* (hemolysin toxin alpha), *sea* (staphylococcal enterotoxin A), *icaA* (intracellular adhesin protein), and *fnbA* (fibronectin-binding protein) under the following PCR conditions: initial denaturation at 94 °C for 5 min, denaturation at 94 °C for 15 s, annealing for 30 s, extension at 72 °C for 35 cycles of 30 s, and a final extension at 72 °C for 45 s. Each reaction consisted of 12.5 µL of AccuPrime master mix (Invitrogen, Waltham, Massachusetts, USA), 1 µL (10 pmol) of the forward primer, 1 µL (10 pmol) of the reverse primer, 2 µL of DNA extract, and the final volume was adjusted to 25 µL with sterile H₂O [29].

Expression of efflux pump and virulence genes

Quantitative real-time PCR (qRT-PCR) was performed to quantify the expression of efflux pump-mediated antibiotic resistance and virulence-associated genes in VRSA isolates. Total RNA was extracted using TRIzol™ reagent (Invitrogen, USA), and RNA integrity was assessed using 1% agarose gel electrophoresis with ethidium bromide staining. The purity and concentration were determined using a NanoDrop™ 2000 spectrophotometer (ThermoFisher Scientific, USA), and only RNA samples with an A260/A280 ratio between 1.8 and 2.1 were used for further analysis.

Genomic DNA contamination was eliminated using TURBO™ DNase (Invitrogen, USA), and its removal was confirmed by the absence of PCR amplification of *gyrB* in a no-reverse transcriptase (no-RT) control reaction. First-strand cDNA synthesis was performed using the RevertAid First Strand cDNA Synthesis Kit (ThermoFisher Scientific, USA) with oligo (dT) primers. TaqMan™ probes and primers were designed using Beacon Designer™ 7.80 (Premier Biosoft, USA) and synthesized by the Eurofins MWG Operon (USA). Primer specificity was verified via BLAST analysis, and dimer formation was evaluated using OligoAnalyzer™ (IDT, USA). Amplification efficiency (90–110%) was confirmed by standard curve analysis using a tenfold serial dilution of cDNA. Multiplex qRT-PCR was performed using the QuantiTect® Multiplex RT-PCR Kit (QIAGEN, Germany) on an ABI 7500 Fast system (Applied Biosystems, France) under the following conditions: 45 °C for 20 min, 95 °C for 15 min, followed by 40 cycles of 94 °C for 1 min and 55 °C for 1 min. Moreover, singleplex reactions were conducted to validate primer efficiency and rule out cross-reactivity before multiplexing. A no-template control (NTC) was included in all runs to confirm the absence of contamination.

To ensure assay reliability, *S. aureus* ATCC 25,923 and UA40 served as positive controls, whereas a gene-negative *S. aureus* strain (lab strain UA1) and NTCs ruled out

nonspecific amplification. The expression stability of the endogenous controls, 16 S rRNA and *gyrB*, was validated using geNorm and NormFinder, with both genes exhibiting M-values < 0.5.

Relative gene expression was calculated using the comparative Ct ($\Delta\Delta C_t$) method, with *S. aureus* ATCC 25,923 as the reference strain (expression level set to 1.0) [30]. A gene was considered significantly overexpressed if it exhibited a ≥ 2.0 -fold increase relative to the reference strain, with statistical significance assessed using Student's t-test ($p < 0.05$). The primer and probe sets used in this study are listed in Supplementary Table 2.

Antimicrobial susceptibility testing

To assess antibiotic susceptibility patterns, all VRSA isolates were subjected to antibiotic sensitivity testing (AST) using the Kirby-Bauer method on Mueller-Hinton agar plates (Oxoid, Basingstoke, Hampshire, UK), following the recommendations of the Clinical Laboratory Standards Institute (CLSI) from 2017 to 2019 [31].

The antibiotic panel included: aminoglycosides (gentamicin 30 µg), ansamycins (rifampicin 5 µg), extended-spectrum cephalosporins (ceftaroline 30 µg, ceftriaxone 30 µg, and cefoxitin 30 µg), quinolones (moxifloxacin 5 µg, and ciprofloxacin 30 µg), beta-lactams (oxacillin 30 µg), folate pathway inhibitors (trimethoprim-sulphamethoxazole 25 µg), fucidanes (fusidic acid 10 µg), oxazolidinones (linezolid 30 µg), glycopeptides (vancomycin 30 µg, teicoplanin 30 µg, and telavancin 30 µg), glycylcyclines (tigecycline 15 µg), lincosamides (clindamycin 2 µg), lipopeptides (daptomycin 30 µg), macrolides (erythromycin 15 µg), phenicols (chloramphenicol 30 µg), phosphonic acids (fosfomycin 50 µg), streptogramins (quinupristin-dalfopristin 15 µg), and tetracyclines (tetracycline 30 µg, doxycycline 30 µg, and minocycline 30 µg).

All antibiotic discs were obtained from Oxoid (Basingstoke, Hampshire, UK), except for the telavancin and daptomycin discs, which were sourced from Hardy Diagnostics (California, USA).

Testing for minimum inhibitory concentration

A minimum inhibitory concentration (MIC) assay was conducted to phenotypically confirm VRSA bacterial strains. The assay was performed using the microtiter plate method, with vancomycin (Oxoid, Basingstoke, Hampshire, UK) as the primary antimicrobial agent. Mueller Hinton broth (Oxoid, Basingstoke, Hampshire, UK) diluted with the appropriate concentration of vancomycin was used in the assay. The MIC was interpreted according to the breakpoints provided by CLSI guidelines [31].

Phenotypic classification of drug-resistant isolates

To determine antimicrobial resistance (AMR) patterns, isolates were classified as MDR, extensively drug-resistant (XDR), or pan-drug-resistant (PDR) based on the standardized criteria proposed by Magiorakos et al. [32]. Isolates were considered MDR if they exhibited resistance to at least one agent in three or more antimicrobial categories, XDR if they were resistant to all but one or two categories, and PDR if they demonstrated resistance to all tested antimicrobial classes.

Multiple antibiotic resistance index

Antibiograms were used to calculate the Multiple Antibiotic Resistance (MAR) index. MAR is an effective method for determining resistance patterns. When calculated for a specific isolate, the MAR index is defined as the ratio between the number of antibiotics to which the isolate showed resistance and the total number of antibiotics tested against the bacterial isolate [33].

Statistical analysis and interpretation of data

For study design illustration, spatial data for the study area was obtained from the online repository at <https://www.diva-gis.org>. A geographic information system (GIS) map was created using the pre-installed sf library in RStudio (version 4.3.2; RStudio Software Inc., Boston, MA, USA). The study design was explained using cartography via Biorender <https://www.biorender.com>.

Sociodemographic data were entered into Microsoft Access XP (version 2404; Microsoft Inc., Redmond, Washington, USA) and exported for further analysis. Data were imported into the SPSS statistical software (version 16.0; SPSS Inc., Chicago, IL, USA). The chi-squared test was used to compare categorical data. Statistical significance for all analyses was assessed using one-sided p values, with significance set at $p < 0.05$.

We conducted a multivariate logistic regression analysis to examine the sociodemographic and clinical factors (sex, age, background, and infection onset categories) associated with the presence of VRSA in HA-UTI. The dependent variable was the presence of *S. aureus* and VRSA (1 = presence, 0 = absence), and the independent variables were age, sex, background, and HA-UTI onset. The regression model was fitted using the glm function in R studio (version 4.3.2, RStudio Software Inc., Boston, MA, USA), with the family parameter set to binomial to specify the logistic regression model. For hypotheses positing a specific directional effect, one-sided tests were conducted by calculating the p -values derived from the model's z -values. Data visualization

was performed using boxplots generated with the ggplot2 package.

However, measures of central tendency were computed, and the detection capacity was assessed using the kappa coefficient to compare genotypic methods (PCR) and phenotypic assays (isolation on BHI-V6 medium). Agreements close to 1 were considered statistically significant at $p < 0.05$.

Student's t -test was performed using GraphPad Prism (Version 10, GraphPad Software Inc., California, USA) to compare the frequency and expression of efflux pumps and virulence factor genes associated with the most prevalent glycopeptide resistance-encoding *van* gene. Additionally, a heatmap was generated to display the presence or absence of each gene.

To elucidate the association between efflux pumps and virulence genes in prevalent *van* gene-carrying VRSA isolates, a correlation matrix was computed using the cor library in RStudio (version 4.3.2, RStudio Software Inc., Boston, USA). The correlation matrix was visualized using the Corr plot function.

Additionally, antibiotic susceptibility patterns were analyzed by clustering with proximity enhancement. The permuted data were displayed on a heat map. A single-linkage hierarchical clustering heat map of the antibiotic resistance profiles of the isolates was generated using the XLSTAT software (Addinsoft, New York, USA).

Results

Socio-demographic and clinical attributes of study participants and prevalence of VRSA isolates

Of the 189 patients screened for cardiac diseases, 37 were clinically diagnosed with HAIs. Among them, 54.05% (20/37) were presented with UTIs, none of which were associated with catheters. Of the HA-UTI cases, 75% (15/20) were female, with a median age of 43 years (range, 17–70 years). Participants were stratified into age groups according to the National Institutes of Health (NIH) guidelines, with older adults comprising the largest proportion (45%, of 9/20). Urban residents accounted for 65% (13/20) of HA-UTI cases, and early onset HA-UTI was more common than late-onset HA-UTI (Table 1A).

All HA-UTI cases were culture-confirmed and yielded a single bacterial isolate per specimen. Among these, 65% (13/20) were *S. aureus*, and 55% (11/13) were phenotypically identified as VRSA. The highest prevalence of *S. aureus* and VRSA was observed among older females (35%, 7/20 and 25%, 5/20, respectively) residing in urban areas (45%, 9/20) with early onset HA-UTI (40%, 8/20). In contrast, young and middle-aged males from rural areas

Table 1 Prevalence of culture-confirmed bacterial isolates stratified by sociodemographic characteristics of study participants. The prevalence of culture-confirmed HA-UTI and VRSA-associated HA-UTI was calculated as the percentage of the total number of isolates (number of isolates/totals $\times$ 100). Key: age group categories: young adults (17–30 years), middle-aged adults (31–45 years), and older adults (46–70 years). **Abbreviations:** HA-UTI: Hospital-acquired urinary tract infection; VRSA: Vancomycin-resistant *S. aureus*; NIH, National institutes of health

Socio-demographic and clinical features		(A)		(B)	
		Prevalence of culture-confirmed HA-UTI		Prevalence of culture-confirmed HA-UTI due to VRSA	
		Frequency (N)	Positive culture (%)	<i>S. aureus</i> N (%)	VRSA N (%)
Gender	Female	15	75	9(45)	9(45)
	Male	5	25	4(20)	2(10)
Age	Young Adults	3	15	1(5)	1(5)
	Middle-aged Adults	8	40	5(25)	5(25)
	Older adults	9	45	7(35)	5(25)
Background	Urban	13	65	9(45)	9(45)
	Rural	7	35	4(20)	2(10)
Onset of infection	Early	13	65	8(40)	8(40)
	Late	7	35	5(25)	3(15)
Specimen	Urine	20	100	13(65)	11(55)
Infection	HA-UTI	20	100	13(65)	11(55)
Ward	CW	20	100	13(65)	11(55)

Abbreviations: HA-UTI: Hospital-acquired urinary tract infection; VRSA: Vancomycin-resistant *S. aureus*; NIH, National institutes of health

exhibited lower prevalence rates (5–25% and 1–5/20 for *S. aureus* and VRSA, respectively), particularly in cases of late-onset HA-UTI ($p < 0.05$) (Table 1B).

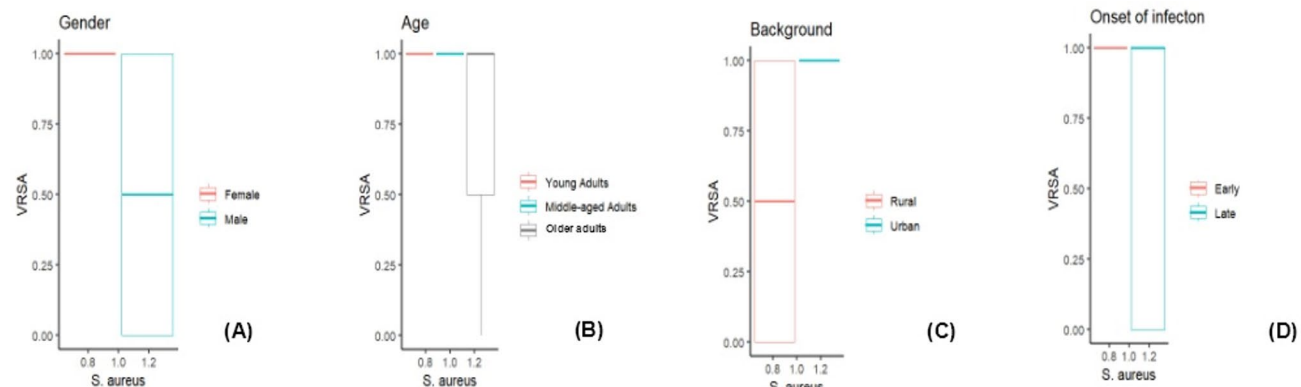


Fig. 2 Demographic and Clinical Factors Associated with VRSA in HA-UTI Isolates. The association of sociodemographic and clinical factors with culture-positive VRSA in HA-UTI isolates was determined using a multivariate logistic regression model. **(A)** Association between sex and culture-positive VRSA in HA-UTI isolates. **(B)** Association between age and culture-positive VRSA in HA-UTI isolates.

Factors influencing culture-confirmed VRSA isolates in patients with HA-UTI

Multivariate logistic regression analysis revealed a potential sex-based (male and female) difference in the presence of VRSA among individuals infected with *S. aureus*. Specifically, females had a higher likelihood of VRSA than males (OR 1, 95% CI, $p < 0.05$) (Fig. 2A).

When analyzing the age categories (young, middle-aged, and older adults), we consistently observed an association between young and middle-aged adults and VRSA presence, whereas older individuals showed more variability (OR 1, 95% CI, $p < 0.05$) (Fig. 2B).

Furthermore, the box plot indicates that the background category (rural or urban) significantly influenced the VRSA levels. Urban areas had a consistently higher likelihood of VRSA, whereas rural areas had a wider range of VRSA values. This suggests that VRSA levels are consistently high in urban settings but more variable in rural settings (OR 1, 95% CI, $p < 0.05$) (Fig. 2C).

Additionally, a multivariate regression plot focusing on the onset of HA-UTI (early and late) and its impact on VRSA and *S. aureus* revealed that early onset infections are consistently associated with high VRSA levels. In contrast, late-onset infections displayed variable VRSA levels, ranging from low to high, indicating a broader spectrum of VRSA in late-onset infections (OR 1, 95% CI, $p < 0.05$) (Fig. 2D).

Overall, these results suggest that females, individuals in the young and middle-aged groups from urban areas, and early onset HA-UTI are associated with VRSA isolates.

(C) Association between background category and culture-positive VRSA in HA-UTI isolates. **(D)** Association between the onset of infection and culture-positive VRSA in HA-UTI isolates. Key: Gender categories include males and females; age categories include young, middle-aged, and older adults. Abbreviations: HA-UTI: Hospital-acquired urinary tract infection; VRSA: Vancomycin-resistant *S. aureus*

Diversity of vancomycin-resistant glycopeptide-encoding Van genes among VRSA isolates in HA-UTI

Singleplex PCR amplification of the *van* gene among phenotypically confirmed VRSA isolates in HA-UTI revealed that all phenotypically identified VRSA isolates (11/11, 100%) were also confirmed through molecular identification. This indicates a consistent agreement between phenotypic and molecular identification methods for VRSA isolates, supported by a high kappa agreement coefficient (1, CI 95% 0.97–1, $p < 0.01$).

Singleplex PCR analysis was conducted to identify the diversity of glycopeptide resistance-encoding genes among all VRSA isolates in HA-UTI by targeting 11 *van* genes (*vanA*, *vanB*, *vanD*, *vanF*, *vanI*, *vanM*, *vanC*, *vanE*, *vanG*, *vanL*, and *vanN*). The results revealed that only *vanA* and *vanB* were detected, with *vanA* showing a higher prevalence (73%, 8/11) than *vanB* (27%, 3/11) ($p < 0.05$) (Table 2A). These results suggest that no other glycopeptide-resistant gene diversity was observed among the HA-UTI VRSA isolates.

Frequency of efflux pump genes and their expression in *vanA*-carrying VRSA isolates from HA-UTI: insights into the resistome

Efflux pump-mediated antibiotic resistance and its prevalence among VRSA isolates from HA-UTI were examined

using singleplex PCR analysis. The analysis revealed that the efflux pump genes *norA* (100%, 8/8), *norB* (88%, 7/8), *norC* (63%, 6/8), and *sepA* (63%, 6/8) were present at the highest frequencies among *vanA*-carrying VRSA isolates (HA-UTI) compared with those carrying *vanB* ($p > 0.05$). Notably, neither *vanA*-nor *vanB*-carrying VRSA isolates in HA-UTI showed the presence of the *mdeA* gene (Table 2B). These results suggest a potential association between the presence of *vanA* and efflux pump genes.

To further investigate this relationship, qRT-PCR analysis was conducted to assess gene expression patterns. The efflux pump genes *norA*, *norB*, *norC*, *sepA*, and *mepA* were significantly upregulated, with fold changes of 11.3, 10.6, 9.8, 10.0, and 4.0, respectively ($p < 0.05$) (Fig. 3A).

Moreover, correlation analysis revealed strong positive associations between *norA* and *norB* ($r = 0.85$, $p < 0.01$), *norB* and *norC* ($r = 0.78$, $p < 0.05$), and *sepA* and *norB* ($r = 0.81$, $p < 0.01$), suggesting a correlation between gene expression levels. In contrast, *mepA* and *norA* were negatively correlated ($r = -0.65$, $p < 0.05$), indicating potential differences in their regulatory mechanisms (Fig. 3B).

These findings underscore the association between *vanA* and efflux pump genes, suggesting a potential link to VRSA resistance phenotypes in HA-UTI, although a direct mechanistic relationship was not demonstrated in this study.

Table 2. Frequency distribution of efflux pumps and virulence genes. The frequency of efflux pumps and virulence genes was calculated as a percentage (number of detected genes/totals $\times 100$) among the most prevalent glycopeptide-resistant encoding *van* genes carrying VRSA isolates in HA-UTI and statistically analyzed

VRSA Isolates in HA-UTI	(A)		(B)										(C)									
	<i>van</i> genes		Efflux Pump genes										Virulome genes									
	<i>vanA</i>	<i>vanB</i>	<i>vanA</i>			<i>vanB</i>				<i>vanA</i>			<i>vanB</i>									
ID	<i>vanA</i>	<i>vanB</i>	<i>norA</i>	<i>norB</i>	<i>norC</i>	<i>sepA</i>	<i>mepA</i>	<i>mdeA</i>	<i>norA</i>	<i>norB</i>	<i>norC</i>	<i>sepA</i>	<i>mepA</i>	<i>mdeA</i>	<i>hla</i>	<i>sea</i>	<i>icaA</i>	<i>fmbA</i>	<i>hla</i>	<i>sea</i>	<i>icaA</i>	<i>fmbA</i>
UA30																						
UA31																						
UA32																						
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51559																						
ATCC																						
51299																						
ATCC																						
29212																						
Frequency %	73	27	100	88	63	63	25	0	67	67	33	33	33	0	50	50	25	63	0	33	33	33
<i>P</i> < 0.05	*								NS									NS				

Key: Light green indicates the absence of genes. Light violet indicates the presence or detection of genes. *Results were considered statistically significant at $p < 0.05$. NS: Non-significant statistical results. Abbreviations: HA-UTI: Hospital-acquired urinary tract infection; VRSA: Vancomycin-resistant *S. aureus*

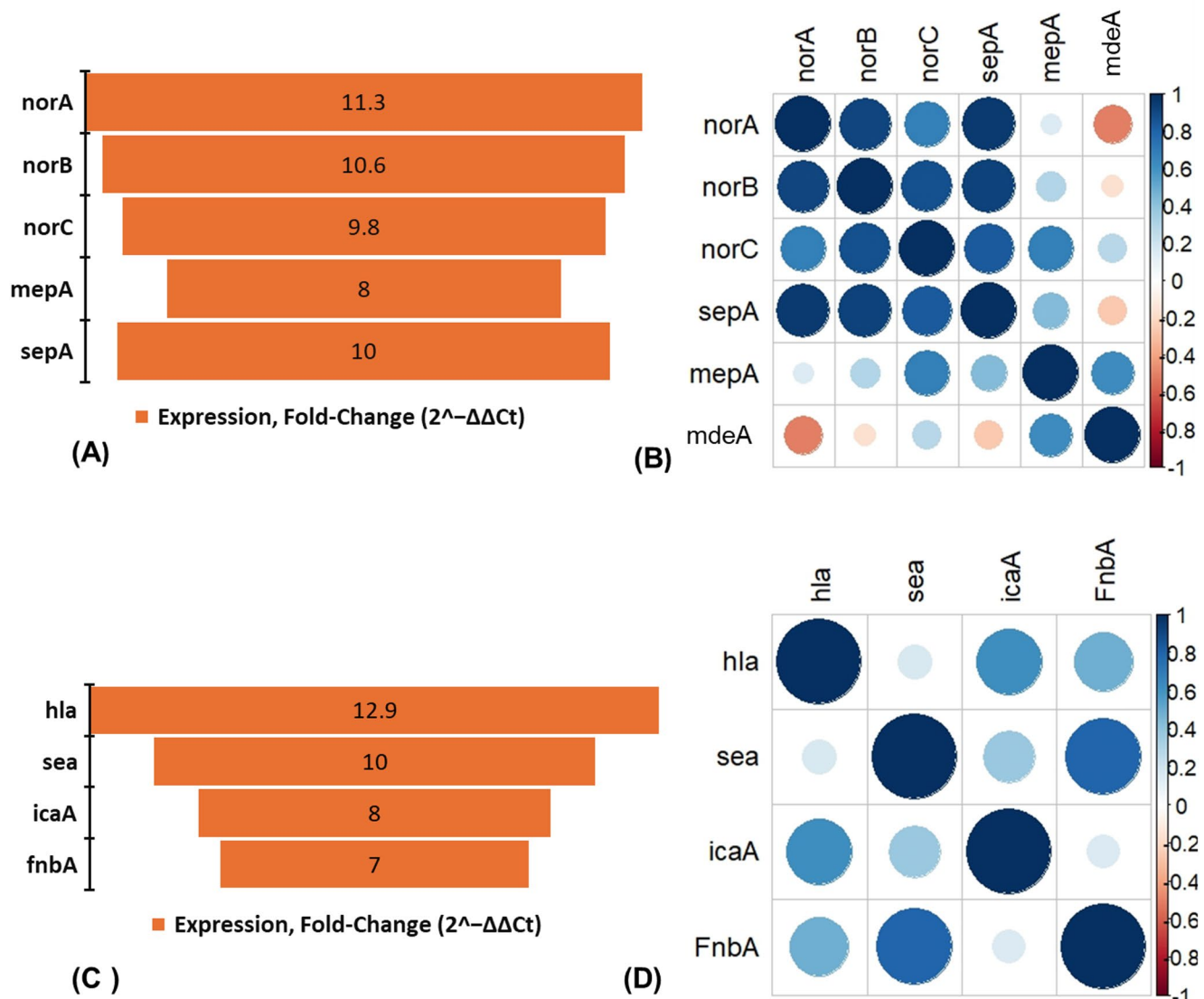


Fig. 3 Expression and Correlation of Efflux Pump and Virulence-Associated Genes in *vanA*-carrying VRSA isolates from HA-UTI. The expression and relationship between efflux pumps and virulence-associated genes in *vanA*-carrying VRSA isolates from HA-UTI were analyzed using qRT-PCR and correlation matrices. **(A)** Efflux pump gene expression: Relative expression levels of *norA*, *norB*, *norC*, *sepA*, and *mepA* were quantified using qRT-PCR. Gene expression was determined by the $2^{\Delta\Delta Ct}$ method, with fold-change values represented on the bars. **(B)** Efflux pump gene correlations: A correlation matrix

illustrating the associations among efflux pump genes in *vanA*-carrying VRSA isolates. **(C)** Virulence gene expression: The expression of *hla*, *sea*, *icaA*, and *fnbA* was quantified using qRT-PCR and expressed as fold-change relative to the control via the $2^{\Delta\Delta Ct}$ method. **(D)** Virulence gene correlations: A correlation matrix depicting associations among virulence genes in *vanA*-carrying VRSA isolates. Abbreviations: HA-UTI, hospital-acquired urinary tract infection; VRSA, vancomycin-resistant *S. aureus*; and qRT-PCR, quantitative real time PCR

Distribution and co-expression of virulence genes in *vanA*-carrying VRSA isolates from HA-UTI: unraveling pathogenicity

The pathogenicity and severity of HA-UTI caused by VRSA isolates were determined by targeting the virulence-encoding genes using singleplex PCR. The results revealed that the prevalence of *fnbA* (63%, 6/8), *hla* (50%, 4/4), and *sea* (50%, 4/4) was the highest among *vanA*-carrying VRSA

isolates in HA-UTI compared with those carrying *vanB* ($p > 0.05$). Notably, the *vanB*-carrying VRSA isolates in the HA-UTI group did not exhibit the presence of *hla*. These findings suggest a potentially greater pathogenic capacity of *vanA*-carrying VRSA isolates in HA-UTI (Table 2C).

To elucidate the pathogenicity and interplay of virulence genes in *vanA*-carrying VRSA isolates, qRT-PCR analysis revealed distinct expression patterns and significant associations ($p < 0.05$) (Fig. 3C). Among the virulence genes, *hla*

and *sea* exhibited the highest expression levels, with fold changes of 12.9 and 10.0, respectively, and a strong positive correlation ($r=0.8$, $p<0.05$), indicating that these genes were co-expressed in our dataset, although the underlying causal mechanisms remain to be elucidated. Similarly, *icaA* and *fnbA* were significantly correlated ($r=0.8$, $p<0.05$), with fold changes of 8.0 and 7.0, respectively, indicating a statistical association between adhesion and biofilm-related gene expression, without confirming a direct functional link (Fig. 3D). Notably, the absence of negative correlations suggests the concurrent upregulation of these virulence genes, which may enhance bacterial persistence and pathogenicity in HA-UTI patients.

Comprehensive antimicrobial susceptibility patterns of VRSA isolates in HA-UTI

A comprehensive antibiogram assessment includes evaluating antibiotic susceptibility, employing hierarchical clustering to identify resistance patterns, determining MICs, classifying drug-resistant phenotypes, and calculating the MAR index for the resistome analysis.

Antibiotic susceptibility testing revealed that 54.5% (6/11) of the *vanA*-carrying VRSA (HA-UTI) isolates were completely resistant to rifampicin, ceftaroline, ceftriaxone, cefoxitin, moxifloxacin, ciprofloxacin, oxacillin, vancomycin, teicoplanin, telavancin, tigecycline, clindamycin, daptomycin, and erythromycin (100%, 11/11). In contrast, 27.2% (3/11) of the *vanB*-carrying VRSA (HA-UTI) isolates were fully susceptible to trimethoprim-sulfamethoxazole, linezolid, clindamycin, chloramphenicol, fosfomycin, and quinupristin-dalfopristin. Notably, all VRSA (HA-UTI) isolates were susceptible to linezolid and chloramphenicol ($p<0.05$) (Fig. 4A). These findings indicated that VRSA in HA-UTI is resistant to multiple antibiotics. However, linezolid and chloramphenicol are viable alternative treatment options.

To analyze antibiotic susceptibility and similarity patterns, hierarchical clustering was performed to generate a heatmap that categorized VRSA (HA-UTI) isolates into three distinct clusters (A, B, and C) based on their resistance profile. Cluster A consisted of three isolates (UA35, UA34, and UA36), Cluster B included five isolates (UA39, UA33, UA31, UA30, and UA32), and Cluster C comprised three isolates (UA37, UA40, and UA38). These clusters demonstrated significant similarities in antibiotic susceptibility ($p<0.05$), suggesting specific patterns of resistance and susceptibility. Notably, this analysis highlighted the effectiveness of linezolid and chloramphenicol as the most effective treatment options against HA-UTI caused by VRSA isolates (Fig. 3B).

Moreover, comparison of antimicrobial susceptibility among *vanA*- and *vanB*-carrying VRSA (HA-UTI) isolates revealed similar resistance rates to ceftaroline, ceftriaxone, cefoxitin, moxifloxacin, ciprofloxacin, oxacillin, vancomycin, teicoplanin, and erythromycin (100%, 11/11) ($p>0.05$). However, the *vanA*-carrying isolates exhibited higher resistance (66.6%) to telavancin, tigecycline, and clindamycin than the *vanB*-carrying isolates ($p<0.05$) (Fig. 3C). Additionally, *vanA* isolates demonstrated higher MIC values (>256 µg/ml) for vancomycin than *vanB* isolates ($p<0.05$) (Table 3A).

Furthermore, all isolates were classified as MDR according to the criteria defined by Magiorakos et al., and no XDR or PDR strains were detected. Notably, 54.5% (6/11) of the *vanA*-positive VRSA isolates exhibited a high MAR index (0.79), which was significantly higher than that of the *vanB*-positive isolates (0.69), highlighting the enhanced resistance profile associated with *vanA*-mediated VRSA ($p<0.05$) (Table 3B).

Discussion

UTIs represent a significant healthcare burden owing to their frequent association with HAIs [34]. This study focused specifically on UTIs caused by VRSA in preoperative patients with extended hospital stays in CW. While prior research has concentrated on UTIs linked to gram-negative bacteria in intensive care units (ICUs) and surgical wards (SWs) (Fig. 1), our investigation uniquely addressed VRSA-related HA-UTI, which pose a distinct and escalating clinical challenge.

The clinical implications of VRSA-related HA-UTI are profound, leading to prolonged hospitalizations, increased healthcare costs, higher rates of treatment failure, limited therapeutic options, and severe complications, such as acute pyelonephritis and bacteraemia [35]. Vulnerable populations, particularly cardiac patients, are particularly susceptible to these infections [36]. Among the 37 participants in this study, 55% were positive for VRSA among patients with HA-UTI. This finding is consistent with the global prevalence trends reported in countries such as Iran, Egypt, and Indonesia, where the incidence of VRSA-related infections in hospital settings continues to rise, particularly in ICUs and cardiac care units, and is often exacerbated by high vancomycin use [37] (Table 1A). Interestingly, studies from Bangladesh and Nepal have reported vancomycin resistance rates of 8% and 11%, respectively, in *S. aureus* isolates [38], whereas our cohort's higher prevalence of 55% underscores the growing challenge of AMR in the Pakistani healthcare setting. These regional variations highlight

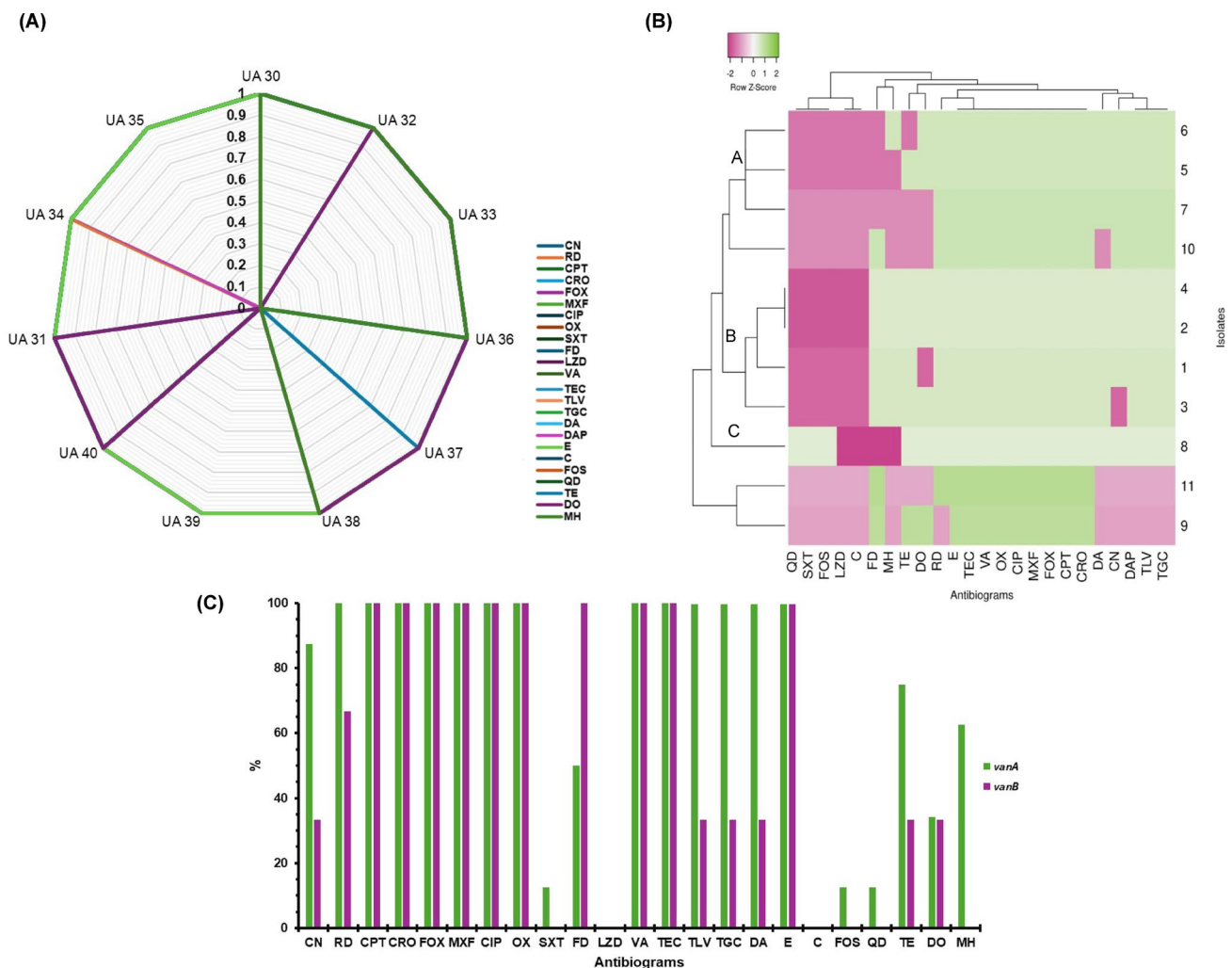


Fig. 4 Antibiotic susceptibility profiles of *vanA* and *vanB* carrying VRSA Isolates. **(A)** Proportional Antimicrobial Resistance (AMR) Breakdown: The frequency of AMR among VRSA isolates from hospital-acquired urinary tract infections (HA-UTI) was represented using a radar graph to illustrate the overall resistance pattern. Antibiotics tested include gentamicin (GN, 30 µg), rifampicin (RD, 5 µg), cefotaxime (CPT, 30 µg), ceftriaxone (CRO, 30 µg), ceftazidime (FOX, 30 µg), moxifloxacin (MXF, 5 µg), ciprofloxacin (CIP, 30 µg), oxacillin (OX, 30 µg), trimethoprim-sulphamethoxazole (SXT, 25 µg), fusidic acid (FD, 10 µg), linezolid (LZD, 30 µg), vancomycin (VA, 30 µg), teicoplanin (TEC, 30 µg), telavancin (TLV, 30 µg), tigecycline (TGC, 15 µg), clindamycin (DA, 2 µg), daptomycin (DAP, 30 µg), erythromycin (E, 15 µg), chloramphenicol (C, 30 µg), fosfomycin (FOS, 50 µg), quinupristin-dalfopristin (QD, 15 µg), tetracycline (TE, 30 µg), doxycycline (DO, 30 µg), and minocycline (MN, 30 µg). The AMR

percentage was calculated as (number of isolates resistant to each antibiotic/total number of isolates) × 100. **(B)** Hierarchical Clustering of Antibiotic Resistance Profiles: A heatmap illustrating the hierarchical clustering of the resistance profiles among VRSA strains from HA-UTI. Single linkage clustering was performed to analyze the resistance patterns and strain similarities using a heatmap generated using XLSTAT. Resistance profiles were displayed horizontally, whereas VRSA isolates were listed vertically, revealing three primary clusters (A, B, and C). **(C)** AMR Variability Between *vanA* and *vanB* carrying VRSA Strains: A bar graph compares the AMR percentages between *vanA* and *vanB* positive VRSA strains, highlighting resistance pattern variations. Isolate key: 1 (UA30), 2 (UA31), 3 (UA32), 4 (UA33), 5 (UA34), 6 (UA35), 7 (UA36), 8 (UA37), 9 (UA38), 10 (UA39), and 11 (UA40)

the need for region-specific strategies to manage VRSA and curb the spread of resistance.

To identify the factors driving VRSA-related HA-UTI and establish associations with infection risk and resistance patterns in bacterial pathogens, we selected demographic variables such as age, sex, and urban/rural background for statistical analysis using a regression model (Table 1B). For instance, UTIs are notably more prevalent in elderly

urban women, likely due to age-related immune decline and environmental factors [39, 40]. In contrast, our study found a higher prevalence of VRSA isolates in younger and middle-aged individuals, underscoring the importance of additional risk factors such as prolonged hospitalization, extended vancomycin treatment, colonization with *Enterococcus* and MRSA, diabetes, chronic skin ulcers, and hemodialysis [41] (Fig. 2). These findings not only highlight the

Table 3. Vancomycin MIC and multiple antibiotic resistance indices. **(A)** Vancomycin MIC was determined using the microtiter plate method for VRSA isolates with HA-UTI. **(B)** To evaluate the extent of antimicrobial resistance (AMR), the multiple antibiotic resistance (MAR) index was calculated as a ratio (number of antibiotics to which the isolate was resistant/total number of antibiotics evaluated). MAR index values were categorized into four distinct ranges: 0.50–0.69, 0.70–0.79, 0.80–0.89, and 0.90–1.00. Isolates highlighted in light green did not fall within these predefined MAR index ratio categories, whereas violet indicated VRSA isolates from HA-UTI with an MAR index ratio

(A)			(B)			
Isolates ID	Vancomycin resistance encoding genes	MIC of vancomycin (µg/mL)	Multiple antibiotic resistant (MAR) index			
			0.5 - 0.69	0.7-0.79	0.8-0.89	0.9-1
UA 30	<i>vanA</i>	>256				
UA 32	<i>vanA</i>	>256				
UA 33	<i>vanA</i>	>256				
UA 36	<i>vanA</i>	>256				
UA 37	<i>vanA</i>	>256				
UA 38	<i>vanA</i>	>256				
UA 39	<i>vanA</i>	>256				
UA 40	<i>vanA</i>	256				
UA 31	<i>vanB</i>	256				
UA 34	<i>vanB</i>	256				
UA 35	<i>vanB</i>	256				

Abbreviations: HA-UTI: Hospital-acquired urinary tract infections; VRSA: Vancomycin-resistant *S. aureus*; MIC: Minimum inhibitory concentration

epidemiological landscape but also emphasize the need for targeted prevention and management strategies that account for both patient demographics and clinical risk factors. In targeted prevention, the role of the hospital environment cannot be ignored, as it serves as a persistent reservoir for VRSA, often harbouring biocide-resistance genes [26]. A recent study demonstrated that an innovative peptide-based disinfectant could eradicate VRSA from hospital surfaces, underscoring the potential of such approaches to complement conventional infection-control measures [21]. This is particularly relevant because the hospital environment and patients are intrinsically interconnected in the dissemination of antimicrobial resistance and pathogenic microbes.

Furthermore, this study also underscores the clinical value of molecular diagnostics, which demonstrated a high level of concordance with phenotypic testing, as indicated by the kappa coefficient. The consistency between molecular methods and traditional phenotypic tests further supports the reliability of molecular diagnostics for detecting VRSA, particularly in settings where the timely and accurate identification of resistant strains is crucial for appropriate therapeutic management [42]. Given the high degree of agreement between phenotypic and molecular methods, the integration of molecular diagnostics into routine clinical practice could significantly improve early detection, reduce the spread of resistance, and enhance treatment outcomes.

The predominance of *vanA* in the VRSA isolates in this study has substantial clinical implications. The presence of

vanA, often located on the mobile transposon Tn1546, facilitates the horizontal transfer of resistance genes, particularly from vancomycin-resistant *Enterococci* to *S. aureus* [43]. This mechanism accelerates the dissemination of drug resistance, particularly in healthcare settings.

Moreover, we examined the roles of efflux pumps and virulence factor genes in VRSA isolates, which contribute significantly to their resistance and pathogenicity [44]. The efflux pump genes *norA*, *norB*, *norC*, and *sepA*, identified in *vanA* carrying VRSA isolates, are known in other contexts to contribute to MDR [45]; in our dataset, their presence was statistically associated with broader resistance profiles (Table 2B). Their co-occurrence, particularly through major facilitator superfamily (MFS) pumps, exacerbates the resistance profiles of these isolates and complicates treatment strategies (Fig. 3A) [45]. Additionally, the co-occurrence of virulence factors such as *fnbA*, *hla*, and *sea* in *vanA*-carrying isolates was associated with markers of pathogenic potential. While these traits are linked to severe infections in other settings, our study did not establish a direct causal role in disease severity (Table 2C and Fig. 3B).

Although clinical outcomes were not assessed, the synergistic upregulation of *hla* and *sea*, potent cytotoxins, together with *icaA* and *fnbA*, which mediate adhesion and biofilm formation, defines a virulence architecture with a clear pathogenic potential [46]. These gene combinations intensify tissue damage, promote immune evasion, and reduce the efficacy of antibiotics. In the context of *vanA*-mediated

resistance, this synergy plausibly increases the risk of persistent, severe, and treatment-refractory HA-UTIs in cardiac patients, underscoring the importance of integrating molecular surveillance with early, targeted interventions.

The integration of these virulence traits into a highly resistant genetic framework amplifies both disease severity and therapeutic complexity, underscoring the need for precision-guided management informed by comprehensive antimicrobial susceptibility testing to contain VRSA and curb its dissemination.

Antibiotic susceptibility testing confirmed that VRSA isolates exhibit high resistance to multiple antibiotics, with significant variability between *vanA*- and *vanB*-carrying isolates [47] (Fig. 4A). VRSA strains harbouring *vanA* exhibited broader resistance profiles, whereas those carrying *vanB* remained susceptible to selected antibiotics, including linezolid, clindamycin, chloramphenicol, fosfomycin, and quinupristin-dalfopristin (Fig. 4B and C).

Moreover, the dominance of *vanA*-mediated resistance in VRSA isolates not only leads to high vancomycin MICs but also contributes to the emergence of MDR strains with broader resistance profiles. In our study, the high MICs (256–512 µg/mL) of vancomycin in *vanA*-carrying isolates [48], combined with a high MAR index (0.79), underscored the growing threat posed by these strains (Table 3B).

This is consistent with global trends, where the predominance of *vanA*-carrying VRSA strains is closely linked to the spread of resistance in both hospital and community settings [26, 49]. The increasing prevalence of these strains highlights the need for rigorous surveillance and the development of alternative antimicrobial agents or combination therapies to manage VRSA infections in the future.

This study also contributes to the limited body of literature on VRSA-associated HA-UTI in patients with cardiac diseases. By investigating the diversity of glycopeptide-encoding *van* genes and their association with efflux pumps, virulence factors, and antibiotic susceptibility, our findings highlight the urgent need for comprehensive strategies that integrate molecular diagnostics, targeted therapies, and robust infection control measures to address this growing public health challenge.

We acknowledge that the single-center design of our study introduced potential regional bias, limiting the generalizability of the findings to diverse healthcare settings. Multicenter investigations are essential for validating these findings and assessing their broader applicability.

The small sample size, which reflects the rarity of VRSA-associated HA-UTI, further reduced the statistical power and weakened the robustness of our conclusions. Additionally, the exclusion of patients on antibiotics and those with recurrent UTIs may have introduced a selection bias,

as these groups often exhibit distinct resistance profiles and clinical dynamics.

The absence of genomic analyses, including *mecA* typing, sequence typing (ST), multilocus sequence typing (MLST), and mobile genetic element (MGE) investigations, limits our understanding of the genetic and mechanistic foundations of resistance, virulence, and HGT. Furthermore, the lack of genetic context analysis for *vanA/vanB* constrains the conclusions regarding HGT.

Moreover, the absence of longitudinal clinical outcome data prevented the direct confirmation of whether the observed co-expression of virulence genes translated into greater infection severity, a prolonged disease course, or a poorer patient prognosis.

Addressing these limitations through multicenter studies with larger, more inclusive cohorts, and comprehensive genomic investigations, including whole-genome sequencing and analysis of environmental and clinical reservoirs, will be essential. Incorporating previously excluded patient groups and capturing longitudinal clinical outcomes will further enhance the applicability of our findings and deepen our understanding of VRSA epidemiology, resistance evolution, and virulence in HA-UTI.

Conclusion

This study found limited diversity in *van* genes among the VRSA isolates, with *vanA* being predominant. The high resistance and pathogenicity observed in HA-UTI isolates were statistically associated with the presence of *vanA*, alongside co-occurring efflux pump genes (*norA+norB*, *norB+norC*, and *sepA+norB*) and virulence determinants (*hla+sea* and *icaA+fnbA*), without implying direct causation.

Despite their MDR profile, these isolates remained susceptible to linezolid and chloramphenicol, highlighting the importance of tailored treatments based on susceptibility testing.

This study represents the first report from Pakistan on the diversity of *van* genes and their association with VRSA-linked HA-UTIs. Understanding the resistome and virulence profiles of VRSA is critical for developing targeted treatment strategies, strengthening infection control measures, and implementing robust antibiotic stewardship programs, particularly in cardiac patients with HA-UTIs.

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Author contributions Muhammad Umer Asghar: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Validation, Visualization, Writing – original draft, Writ-

ing – review & editing. Muhammad Tariq: Supervision. Arsalan Has-
eeb Zaidi: Supervision. Kabeer Haneef: Writing, review, and editing.
Aisha Asghar: Writing - review & editing. Noor Ul Ain: Investigation,
Writing, review, and editing.

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Data availability All data analyzed in this study were obtained from
the corresponding author upon request.

Declarations

Consent for publication All the authors have reviewed and approved
the final manuscript.

Competing interests The authors declare no competing interests.

Ethical approval This study was conducted in accordance with the eth-
ical principles of the Declaration of Helsinki. The direct involvement
of patients was not included because only medical data were used. This
study was approved by the Institutional Ethical Review Committee of
the Faisalabad Institute of Cardiology (FIC) (approval no. 17–2019/
DME/FIC/FSD). As this study involved retrospective medical data,
the requirement for informed consent was waived by the Institutional
Ethical Review Committee of the FIC.

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